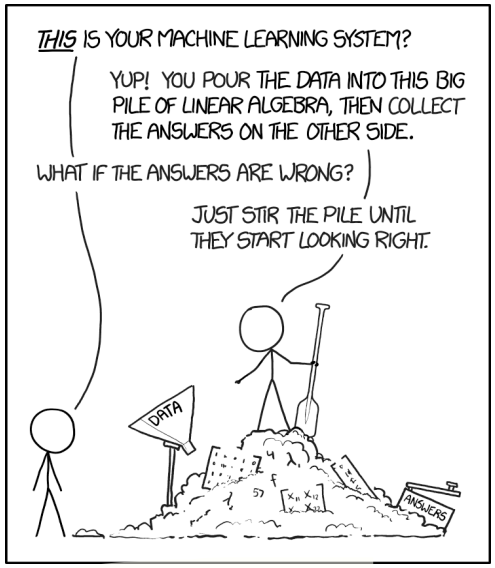
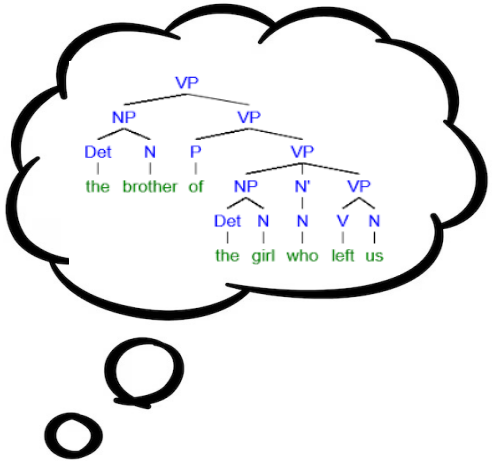
Can (perhaps future) AI **think**?

What **is** thinking?





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Agenda

- The Computational Theory of Mind (CTM) ≈ 'Machine Functionalism', including the Language of Thought Hypothesis (LOT).
    - 'reason-respecting behavior'
    - syntax & semantics
    - derived & underived meaning
    - productivity, systematicity, compositionality.
- The challenge from Connectionism (against necessity)
- The Chinese Room argument (against sufficency)

+ responses

What is thinking (in humans)?

CTM: formal symbol manipulation.

Then why *couldn't* AI think?

Functionalism (again)

Compare:

'What's a *feed dog*?

'It's the thing that does that thing in this thing!'

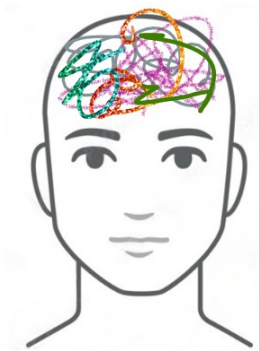
'What thing in which thing?'

(sewing machine)

'What's a *pain*?'

'It's the state that does this thing in that thing.'

'Which thing in what thing?'



(human system)

So let's get a little more precise.

Toy **sewing machine theory**:

> "… The drop feed mechanism is used by almost all household machines and involves a mechanism below the sewing surface of the machine. When the needle is withdrawn from the material being sewn, a set of feed dogs is pushed up through slots in the machine surface, then dragged horizontally past the needle. The dogs are serrated to grip the material, and a presser foot is used to keep the material in contact with the dogs. At the end of their horizontal motion, the dogs are lowered again and returned to their original position." (Etc. etc.)

(Source: https://en.wikipedia.org/wiki/Sewing_machine)

'Ramsey sentence' for toy sewing machine theory:

> $\exists\, x_1, x_2, x_3$ (… When the x_1 is withdrawn from the material being sewn, a set of x_2 is pushed through slots in the machine surface… and an x_3 is used to keep the material in contact with the x_2…)

Functional definitions ('needle' & 'feed dog'):

> *needle* = an x_1 such that $\exists\, x_2, x_3$ (… When the x_1 is withdrawn from the material being sewn, a set of x_2 is pushed through slots in the machine surface…)
>
> *feed dog* = an x_2 such that $\exists\, x_1, x_3$ (… When the x_1 is withdrawn from the material being sewn, a set of x_2 is pushed through slots in the machine surface…)

Toy **psychological theory**:

"… and pain is typically caused by tissue damage, and typically causes worry, sadness and screaming, and worry and sadness typically cause avoidance behavior… (etc.)… and hunger is typically caused by an empty stomach, and typically causes food seeking behavior…" (Etc. etc.)

Ramsey sentence for toy psychology:

$\exists\, x_1, x_2, x_3$ (… and x_1 is typically caused by tissue damage, and typically causes x_2 and x_3 and screaming, and x_2 and x_3 in turn typically cause avoidance behavior…")

Functional definitions (of "pain" & "worry")

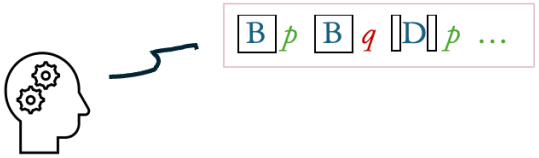
pain = the x_1 such that $\exists\, x_2, x_3$ (… and x_1 typically causes x_2 and x_3 and screaming, and x_2 and x_3 in turn typically cause avoidance behavior…)

worry = the x_2 such that $\exists\, x_1, x_2$ (… and x_1 typically causes x_2 and x_3 and screaming, and x_2 and x_3 in turn typically cause avoidance behavior…)

N.B. emphasis on *toy* (theory)..

Remember the big selling point of functionalism: **multiply realizable** (both sewing machines and minds)

Back to **thinking**



To worry isn't to think, and a pain isn't a thought..

Examples:

- coming to believe that it's raining because one believes that the streets are wet
- deciding to take the later train because one wants to do some shopping in Hamar (and believes that the early train won't allow for that)
- reasoning from 'it's Friday', and 'if it's Friday there's no class tomorrow', to 'there's no class tomorrow'

the belief that
(or perhaps the wish that..)

the belief that

the belief that

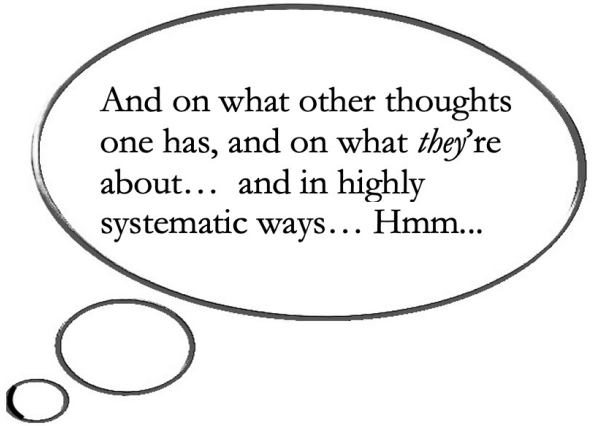
p
$p > q$
---
q

Beliefs — and other so-called 'propositional attitudes', like desires and intentions — have (propositional) **contents**: e.g., that it's raining, that there's no class tomorrow.

propositional content: content that can be specified this way ('that p')

Crucially, thoughts have *contents* — they're about something.

And what a thought 'does' depend in large part on what it's about.



What's a **computer program**?

At a very abstract level: a set of specified operations on some input (and, perhaps, internal states) that yield a certain output.

More specifically: at least some of the input and output states (and perhaps, internal states) are **representational** — they have meaning/semantic content — and/or can be 'usefully interpreted as' being representational.

But computer programs aren't explicitly representational — they don't manipulate semantic features **directly**. They manipulate semantics *by* manipulating **syntax**.

>>> 'formal symbol manipulation'

Compare: 'rain', 'regn', 'дощ', '11110001', 'Ω', '🌧', '

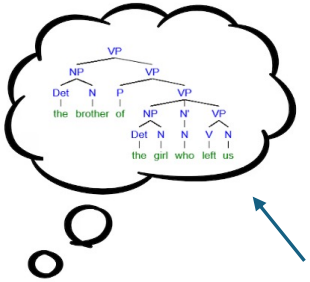
The Computational Theory of Mind

A theory of thinking (computational operations): as formal symbol manipulation.

+

A theory of the 'manipulated states' — beliefs and other propositional attitudes — as functional states: causal relations to input, output and <u>mental representations</u>.

What kind of representations? LOT (early Fodor): 'sentences in the head' ..



N.B. Very misleading illustration..

More carefully, representations with linguistic structure.

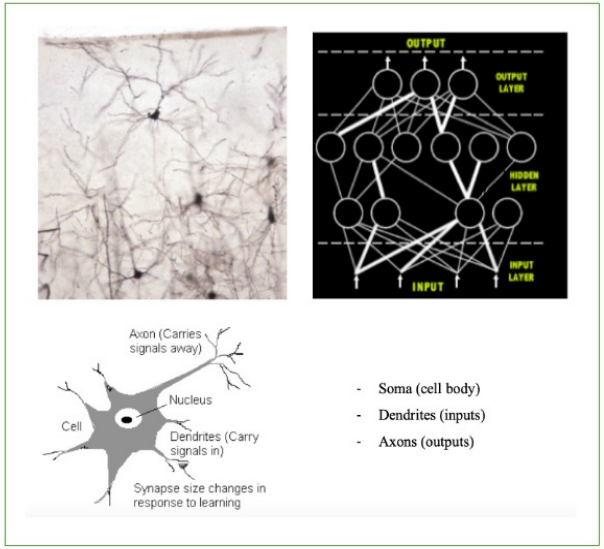
In particular: symbolic representations with a <u>combinatorial syntax</u>, <u>whose semantic properties map onto the syntactic properties</u>.

Arguments for LOT (and CTM):

1. Semantic parallels between language and thought — 'truth-value inheritance'.
2. Syntactic parallels between language and thought — **systematicity** and **productivity**.

Arguments against CTM (and LOT):

1.
2.



Responses include:

Chinese room

- the 'systems reply'
- question-begging?

Connectionism

- compatible? (implementation)
- fails to explain productivity & systematicity

Returning to the opening questions

In what way, if any, is thinking (in human systems) formal symbol manipulation?

Could AI systems think? If not, why not?